Exact Optimal Cloning of Stabilizer States

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Abstract

We determine the optimal average global fidelity of $T \to T+k$ approximate cloning for the uniform ensemble of n-qubit stabilizer states, for every number of qubits n, every $k \geq 1$ and every $T \leq 3$. With $D = 2^n$, the optimum is

$$P_{\text{stab}}(n;T,k) = \binom{D+T-1}{T} F_{T+k-\kappa(k)}, \qquad F_s = \frac{\prod_{i=0}^{s-2}(2^i+1)}{D\prod_{i=0}^{s-2}(D+2^i)},$$

where F_s is the stabilizer frame potential and $\kappa(k)$ is the largest dimension of a doubly-even binary code of length k. The result rests on an exact formula for the largest eigenvalue of the mixed stabilizer moment operators $\mathbb{E}_S\, |S\rangle\langle S|^{\otimes a} \otimes |\bar{S}\rangle\langle\bar{S}|^{\otimes b}$: for all $a+b \geq 1$ and all n it equals F_{a+b-w}, where $w = \min(a,b) + \kappa(|a-b|)$ is the Witt index of an associated $\mathbb{Z}_4$-valued quadratic form. The proof works in the algebra of stochastic Lagrangian relations of Gross, Nezami and Walter, and needs neither a stable-range assumption nor the linear independence of the relation operators. Consequences: the universal cloner is optimal exactly when $T+k \leq 3$; the optimum stays constant as k grows across the lengths at which κ increases; and as $k \to \infty$ it converges to the optimal probability of identifying the state.

1 Introduction

Werner determined the optimal global fidelity of cloning an unknown pure state [1]. For stabilizer states, Herasymenko et al. observe, in a trace-distance-oriented discussion, that the optimal cloning channel remains an open question [2], and Bansal, Caro and Mahajan establish the optimal constant-error cloning sample complexity $\Theta(n)$ [3]. Separately, Jeon, Sohn and Oh identify exact optimal fidelities beyond asymptotic scaling as an open problem for general mixed-state cloning [4]. These results address different figures of merit or parameter regimes from the finite-copy average global fidelity considered here. Since stabilizer states form a complex projective 3-design but not a 4-design [5, 6], Werner's universal cloner is optimal for them exactly when $T+k \leq 3$ [7]. To our knowledge, no general closed formula had been established for the optimal average global fidelity of uniform n-qubit stabilizer-state cloning throughout the non-design regime $T+k \geq 4$, for arbitrary n and all $1 \leq T \leq 3$.

We determine them. For $T \leq 3$, every k and every n, the optimal average global fidelity is the closed form in the abstract (Corollary 11); Table 1 lists examples. The formula has two ingredients: the frame potential F_s of random stabilizer states [8][5, Thm. 2], and the coding-theoretic quantity $\kappa(k)$, the largest dimension of a doubly-even binary code of length k. The appearance of κ has a precise source: it is the Witt index of the $\mathbb{Z}_4$-valued quadratic form that indexes the relevant stabilizer moment.

Table 1: Optimal $T \to T+k$ cloning fidelity of uniform n-qubit stabilizer states (Corollary 11), with Werner's universal value in parentheses.

	$n=1$	$n=2$	$n=3$	$n=4$	$n=5$
$1 \to 4$	5/12 (2/5)	1/8 (4/35)	1/36 (4/165)	1/204 (4/969)	5/6732 (4/6545)
$1 \to 5$	5/12 (1/3)	1/8 (1/14)	1/36 (1/99)	1/204 (1/969)	5/6732 (1/11781)
$1 \to 6$	3/8 (2/7)	3/32 (1/21)	1/64 (2/429)	1/544 (2/6783)	1/5984 (2/145299)
$2 \to 4$	5/8 (3/5)	5/16 (2/7)	1/8 (6/55)	1/24 (2/57)	5/408 (6/595)
$2 \to 5$	9/16 (1/2)	15/64 (5/28)	9/128 (1/22)	1/64 (1/114)	3/1088 (1/714)
$3 \to 4$	5/6 (4/5)	5/8 (4/7)	5/12 (4/11)	1/4 (4/19)	5/36 (4/35)
$3 \to 5$	3/4 (2/3)	15/32 (5/14)	15/64 (5/33)	3/32 (1/19)	1/32 (1/63)

Method. For $T \le 3$, Clifford symmetry reduces cloning to the largest eigenvalue of a mixed moment operator (Section 2). Gross, Nezami and Walter express stabilizer moments as sums over stochastic Lagrangian relations [8]; a partial transpose gives the mixed moments (Section 3). We compute the top eigenvalue for every signature by passing to the abstract algebra of relations, where left multiplication by the moment is a nonnegative matrix whose column sums depend only on a "defect level" and are given exactly by frame potentials (Section 5). The top level is the Witt index, computed in Section 4. Related exact work by Hayashi and Lu diagonalizes the stabilizer Gram operator for state identification [9]; our argument does not use their spectral theorem. The moment identities [8][9, Lemma 6] and the frame potentials are known; what is new here is the top eigenvalue of the mixed moments (Theorem 10) and its consequence for cloning (Corollary 11).

2 Setting and reduction

Let $D = 2^n$, let $\mathrm{Stab}(n)$ be the set of n-qubit stabilizer states, $N_n = |\mathrm{Stab}(n)| = 2^n \prod_{i=1}^{n}(2^i+1)$, and $D_m = \binom{D+m-1}{m} = \dim \mathrm{Sym}^m(\mathbb{C}^D)$. A $T \to T+k$ cloner is a channel Φ from operators on $\mathrm{Sym}^T(\mathbb{C}^D)$ to operators on $\mathrm{Sym}^{T+k}(\mathbb{C}^D)$. This loses no generality: all inputs lie in $\mathrm{Sym}^T(\mathbb{C}^D)$, and projecting the output of an unrestricted channel onto $\mathrm{Sym}^{T+k}(\mathbb{C}^D)$, replacing the complementary outcome by a fixed state in that subspace, gives a channel whose fidelity with every $|S\rangle^{\otimes(T+k)}$ is no smaller. Then

$$P_{\mathrm{stab}}(n;T,k) = \max_{\Phi}\ \mathbb{E}_S \langle S^{\otimes(T+k)} | \Phi\big(|S\rangle\langle S|^{\otimes T}\big) | S^{\otimes(T+k)} \rangle,$$

with S uniform on $\mathrm{Stab}(n)$. For $a, b \ge 0$ define the mixed moment

$$X^{(a,b)} = \mathbb{E}_S\, |S\rangle\langle S|^{\otimes a} \otimes |\bar{S}\rangle\langle \bar{S}|^{\otimes b}.$$

Lemma 1 (Reduction). *For $1 \le T \le 3$ and $k \ge 1$, $P_{\mathrm{stab}}(n;T,k) = D_T\, \lambda_{\max}\big(X^{(T+k,T)}\big)$. Moreover the optimum is attained by a Clifford-covariant channel, which has the same fidelity on every stabilizer state, so the average and worst-case optima coincide.*

Proof. Fix an orthonormal basis $\{e_c\}$ of $\mathrm{Sym}^T(\mathbb{C}^D)$ and write $s_c(S) = \langle e_c | S^{\otimes T}\rangle$. With the Choi operator $J = \sum_{c,c'} \Phi(|e_c\rangle\langle e_{c'}|) \otimes |c\rangle\langle c'|$, the average fidelity is $\mathrm{tr}(JX)$ with $X = \mathbb{E}_S vv^\dagger$, $v = S^{\otimes(T+k)} \otimes \sum_c \overline{s_c(S)}|c\rangle$. This X is the compression of $X^{(T+k,T)}$ to $\mathrm{Sym}^{T+k} \otimes \overline{\mathrm{Sym}^T}$, which contains its support, so the two have the same nonzero spectrum. By semidefinite-programming duality (a strictly feasible point exists), $P_{\mathrm{stab}} = \min\{\mathrm{tr}\, Y : I \otimes Y \succeq X\}$, the minimum being attained. For a Clifford unitary C, X is invariant under $C^{\otimes(T+k)} \otimes \bar{C}^{\otimes T}$, so averaging an optimal Y over the Clifford group gives an optimal Y commuting with the action of $\bar{C}$ on Sym^T. For $T \le 3$, the unitary 3-design property of

the multiqubit Clifford group [10, 11] implies that its twirl on $\mathrm{End}((\mathbb{C}^D)^{\otimes T})$ equals the Haar twirl; an operator commuting with the Clifford action on $\mathrm{Sym}^T(\mathbb{C}^D)$, extended by zero, is therefore fixed by the Haar twirl, and irreducibility of $\mathrm{Sym}^T(\mathbb{C}^D)$ under $\mathrm{U}(D)$ forces it to be scalar. Thus $Y = \lambda I$. The constraint is then $\lambda \geq \lambda_{\max}(X)$, giving $P_{\mathrm{stab}} = D_T \lambda_{\max}(X)$. Averaging an optimal channel over the (transitive) Clifford action gives a covariant optimal channel with equal fidelity on every stabilizer state. □

3 Mixed moments as sums over Lagrangian relations

Write $t = a + b \geq 1$ and $V = \mathbb{F}_2^t$ with coordinates split into A (the first a) and B (the last b). Define the $\mathbb{Z}_4$-valued form $q_{a,b}(x) = |x_A| - |x_B| \bmod 4$, where $|\cdot|$ is the Hamming weight, and the dot product $\beta(x, y) = x \cdot y \bmod 2$. Then $q(x + y) = q(x) + q(y) + 2\beta(x, y) \pmod 4$.

Definition 2. $\Sigma(q_{a,b})$ is the set of t-dimensional subspaces $U \subseteq V \oplus V$ such that $q_{a,b}(x) \equiv q_{a,b}(y) \pmod 4$ for all $(x, y) \in U$, and $\mathbf{1}_{2t} = (\mathbf{1}_t, \mathbf{1}_t) \in U$. For $U \in \Sigma$ let $r(U) = \sum_{(x,y)\in U} |x\rangle\langle y|$ on $(\mathbb{C}^2)^{\otimes t}$, $R(U) = r(U)^{\otimes n}$ (acting on $(\mathbb{C}^D)^{\otimes t}$ after reordering), $N_L(U) = \{x : (x, 0) \in U\}$ and $N_R(U) = \{y : (0, y) \in U\}$.

Every $U \in \Sigma$ is totally isotropic for $q \oplus -q$, hence for the polar form $\beta \oplus \beta$, and has dimension t, so $U = U^\perp$; in particular $\dim N_L(U) = \dim N_R(U)$, and both are totally isotropic for $q_{a,b}$. For $b = 0$, $\Sigma(q_{t,0})$ is the set $\Sigma_{t,t}$ of Gross–Nezami–Walter [8, Def. 4.1].

Lemma 3 (Mixed moments)**.** *For $t = a + b \geq 1$ and $n \geq 1$, $X^{(a,b)} = Z_{n,t}^{-1} \sum_{U\in\Sigma(q_{a,b})} R(U)$ with $Z_{n,t} = D \prod_{i=0}^{t-2}(D + 2^i)$.*

Proof. For $b = 0$ this is [8, Thm. 5.3], valid for all $n, t \geq 1$. In general $|\bar{S}\rangle\langle\bar{S}|$ is the transpose of $|S\rangle\langle S|$, so $X^{(a,b)}$ is the partial transpose of $X^{(t,0)}$ on the last b factors. On a single relation, $r(U)^{\Gamma_B} = \sum_{(x,y)\in U} |x_A y_B\rangle\langle y_A x_B| = r(U')$ with $U' = \{((x_A, y_B), (y_A, x_B))\}$. The map $U \mapsto U'$ preserves dimension and $\mathbf{1}_{2t}$, and $q_{a,b}(x_A, y_B) - q_{a,b}(y_A, x_B) = |x| - |y|$, so it is a bijection $\Sigma_{t,t} \to \Sigma(q_{a,b})$. The same statement appears as [9, Lemma 6]. □

Composition. For $U_1, U_2 \in \Sigma$ let $U_1 \circ U_2 = \{(x, z) : \exists y,\ (x, y) \in U_1, (y, z) \in U_2\}$ and $d(U_1, U_2) = \dim\big(N_R(U_1) \cap N_L(U_2)\big)$.

Lemma 4. *$r(U_1)r(U_2) = 2^{d(U_1,U_2)}\, r(U_1 \circ U_2)$ and $U_1 \circ U_2 \in \Sigma$. Hence $R(U_1)R(U_2) = D^d R(U_1 \circ U_2)$.*

Proof. The coefficient of $|x\rangle\langle z|$ in $r(U_1)r(U_2)$ counts y with $(x, y) \in U_1$, $(y, z) \in U_2$; this set is empty or a coset of $N_R(U_1) \cap N_L(U_2)$. The fibre product $\{(x, y, z)\}$ has dimension $2t - \dim(\pi_2 U_1 + \pi_1 U_2) = 2t - (t - d) = t + d$, using $(\pi_2 U_1)^\perp = N_R(U_1)$ and $(\pi_1 U_2)^\perp = N_L(U_2)$ (from $U = U^\perp$); its projection to $U_1 \circ U_2$ has kernel of dimension d, so $\dim U_1 \circ U_2 = t$. If y witnesses (x, z) then $q(x) - q(z) = (q(x) - q(y)) + (q(y) - q(z)) = 0$, and $\mathbf{1}_{2t} \in U_1 \circ U_2$ because it lies in both. □

Let $\mathcal{A}$ be the algebra with basis $\{e_U\}_{U\in\Sigma}$ and product $e_U e_{U'} = D^{d(U,U')} e_{U\circ U'}$. It has unit e_Δ for the diagonal $\Delta = \{(x, x)\} \in \Sigma$, and by Lemma 4 and bilinearity the map $\varphi(e_U) = R(U)$ satisfies $\varphi(uv) = \varphi(u)\varphi(v)$ and $\varphi(e_\Delta) = I$ (associativity of $\mathcal{A}$ is never used). Put $x = Z^{-1} \sum_U e_U$, so $\varphi(x) = X^{(a,b)}$, and let M be the matrix of left multiplication by x:

$$M_{U'',U'} = Z^{-1} \sum_{U:\ U\circ U' = U''} D^{d(U,U')} \ \geq 0.$$

Lemma 5 (Spectrum). *$\lambda_{\max}(X^{(a,b)}) = \rho(M)$, the spectral radius of M.*

Proof. Since $M^j e_\Delta = x(x(\cdots(xe_\Delta)))$ and φ is multiplicative, $\varphi(M^j e_\Delta) = X^j$; so for the minimal polynomial p of M we get $0 = \varphi(p(M)e_\Delta) = p(X)$; hence every eigenvalue of X is a root of p, i.e. an eigenvalue of M, and $\lambda_{\max}(X) \le \rho(M)$. Conversely, let $y \ge 0$, $y \ne 0$, satisfy $My = \rho(M)y$ (Perron–Frobenius for nonnegative matrices; no irreducibility is needed). Then $X\varphi(y) = \varphi(xy) = \rho(M)\varphi(y)$. Every $U \in \Sigma$ contains $(\mathbf{1}_t, \mathbf{1}_t)$, so $\langle \mathbf{1}_t^{\otimes n} | R(U) | \mathbf{1}_t^{\otimes n} \rangle = 1$ and $\langle \mathbf{1}_t^{\otimes n} | \varphi(y) | \mathbf{1}_t^{\otimes n} \rangle = \sum_U y_U > 0$. Hence $\varphi(y) \ne 0$ and $\rho(M)$ is an eigenvalue of X. □

4 The Witt index of $q_{a,b}$

Let $w(q)$ be the largest dimension of a subspace $W \subseteq V$ with $q|_W = 0$. Doubly-even binary codes of length m are exactly the totally isotropic subspaces of $q_{m,0}$, so $\kappa(m) = w(q_{m,0})$.

Proposition 6. *Let $\mu(j) = 0,1,2,3,2,3,2,1$ for $j \equiv 0,\dots,7 \pmod 8$. Then $w(q_{a,b}) = \big(t - \mu(a-b)\big)/2$. Equivalently $\kappa(m) = 4\lfloor m/8 \rfloor + (0,0,0,0,1,1,2,3)_{m \bmod 8}$ and $w(q_{a,b}) = \min(a,b) + \kappa(|a-b|)$.*

Proof. Upper bound. (i) $0 = q(2x) = 2q(x) + 2\beta(x,x)$, so $q(x) \equiv \beta(x,x) \pmod 2$. (ii) The Gauss sum factorizes over coordinates: $\gamma(q) = \sum_x i^{q(x)} = (1+i)^a (1-i)^b = 2^{t/2}\zeta_8^{a-b}$. (iii) If $q|_W = 0$ then $\beta|_W = 0$; for $x \notin W^\perp$, $\sum_{w \in W} i^{q(x+w)} = i^{q(x)} \sum_w (-1)^{\beta(x,w)} = 0$, while q is constant on cosets of W in $W^\perp$. Hence $\gamma(q) = |W|\gamma(\bar q)$ for the induced form on $W^\perp/W$, which has dimension $m = t - 2\dim W$, a nondegenerate polar form, and $\gamma(\bar q) = 2^{m/2}\zeta_8^{a-b}$. (iv) If $m = 0$ then $\gamma(\bar q) = 1$, forcing $a - b \equiv 0 \pmod 8$; if $m = 1$ the single basis vector has odd $\bar q$ by (i), so $\gamma(\bar q) = 1 \pm i$, forcing $a - b \equiv \pm 1$. With $m \equiv a - b \pmod 2$ this gives $m \ge \mu(a-b)$, i.e. $\dim W \le (t - \mu(a-b))/2$.

Lower bound. The words 1111; 11110; $\{111100, 110011\}$; the $[7,3]$ simplex code; and the extended Hamming $[8,4,4]$ code are doubly even, of dimensions $1,1,2,3,4$ at lengths $4,5,6,7,8$. Direct sums with the $[8,4,4]$ code give $\kappa(m+8) \ge \kappa(m) + 4$, so $\kappa(m) \ge (m - \mu(m))/2$, with equality by the upper bound. For $q_{a,b}$ with $a \ge b$, the b vectors $e_{A_i} + e_{B_i}$ ($q = 1 - 1 = 0$, disjoint supports) together with a maximal doubly-even code on the remaining $a - b$ coordinates of A span a totally isotropic subspace of dimension $b + \kappa(a-b)$; the case $a < b$ follows by exchanging A and B and replacing q by $-q$. Since $\kappa(m) = (m - \mu(m))/2$, this matches the upper bound. □

5 The top eigenvalue

For $s \ge 1$ set $F_s = \prod_{i=0}^{s-2}(2^i+1)/\big(D\prod_{i=0}^{s-2}(D+2^i)\big)$, and write $(D)_k = \prod_{i<k}(D - 2^i)$, $c_k = \prod_{i=0}^{t-k-2}(2^i+1)$, and $\binom{r}{k}_2$ for the Gaussian binomial. The *level* of $U' \in \Sigma$ is $\dim N_L(U')$.

Lemma 7 (Column sums). *The column of M indexed by U' sums to $Z^{-1}\sum_{U\in\Sigma} D^{\dim(N_R(U)\cap N_L(U'))}$, and if U' has level r this equals $Z^{-1}S_t(r)$ with $S_t(r) = \sum_{k=0}^r \binom{r}{k}_2 (D)_k\, c_k$.*

Proof. Each U contributes to exactly one row ($U'' = U \circ U'$), which gives the first expression. For a subspace W, $D^{\dim W} = |W|^n = \sum_{K \le W} (D)_{\dim K}$, by grouping n-tuples of vectors of W according to their span ($(D)_k$ counts surjections $\mathbb{F}_2^n \to K$, and vanishes when $k > n$). Exchanging sums, $\sum_U D^{\dim(N_R(U)\cap K')} = \sum_{K \le K'} (D)_{\dim K} \#\{U : K \le N_R(U)\}$ with $K' = N_L(U')$, and Lemma 8 evaluates the count as $c_{\dim K}$. □

Lemma 8 (Counting). *For every subspace $K \le V$ of dimension k that is totally isotropic for $q_{a,b}$, $\#\{U \in \Sigma : (0 \oplus K) \subseteq U\} = c_k$.*

Proof. Let $J = \langle \mathbf{1}_{2t} \rangle \oplus (0 \oplus K)$. Elements of K have $q = 0$, hence even weight, so $(\beta \oplus \beta)(\mathbf{1}_{2t}, (0,h)) = 0$ for $h \in K$ and J is totally isotropic for $q \oplus -q$, of dimension $k+1$. On $J^\perp \subseteq \mathbf{1}_{2t}^\perp$ the form $q \oplus -q$ takes even values, so $Q = (q \oplus -q)/2 \bmod 2$ is an $\mathbb{F}_2$-quadratic form there with polar form $\beta \oplus \beta$; its radical on $J^\perp$ is J, so Q induces a nondegenerate quadratic form on $J^\perp/J$, of dimension $2m$ with $m = t-1-k$. Elements of Σ containing J correspond bijectively to Lagrangians (maximal totally singular subspaces) of $J^\perp/J$. A Lagrangian exists: $U_K = \{(u,v) : u, v \in K^\perp,\ u+v \in K\}$ has dimension t, contains J (since $\mathbf{1}_t \in K^\perp$), and satisfies $q(u) - q(u+h) = -q(h) - 2\beta(u,h) = 0$ for $h \in K$. A nondegenerate $2m$-dimensional $\mathbb{F}_2$-quadratic space containing a totally singular m-space is an orthogonal sum of m hyperbolic planes (choose e in it and f with $\beta(e,f) = 1$, replace f by $f + Q(f)e$, and induct). In such a space the Lagrangians number $N(m) = \prod_{i=0}^{m-1}(2^i+1)$: with a hyperbolic pair (e_0, f_0) split off, one Lagrangian contains e_0 for each Lagrangian L' of the residual space, and those not containing e_0 are $\{x + \phi(x)e_0 : x \in L'\} \oplus \langle f_0 + w + Q(w)e_0 \rangle$ for a linear functional ϕ on L' and any w in the residual space with $\beta(w,x) = \phi(x)$ for all $x \in L'$ (such w exist by nondegeneracy, and the resulting Lagrangian does not depend on the choice). Thus $N(m) = (1 + 2^{m-1})N(m-1)$, $N(0) = 1$, and the count is $N(t-1-k) = c_k$. □

Lemma 9 (A Gaussian-binomial identity)**.** *$S_t(r) = c_r \prod_{i=t-r-1}^{t-2}(D + 2^i)$. Consequently $Z^{-1}S_t(r) = F_{t-r}$.*

Proof. Put $m = t-r-1$ and $a_j = \prod_{i<j}(2^{m+i}+1)$, so $c_k = c_r\, a_{r-k}$. It suffices to show $L_r := \sum_k \binom{r}{k}_2 (D)_k a_{r-k} = \prod_{i<r}(D + 2^{m+i})$. Using $\binom{r}{k}_2 = 2^k \binom{r-1}{k}_2 + \binom{r-1}{k-1}_2$, $a_{r-k} = a_{r-1-k}(2^{m+r-1-k}+1)$ and $(D)_k = (D)_{k-1}(D - 2^{k-1})$, the two contributions to the coefficient of $\binom{r-1}{j}_2 (D)_j a_{r-1-j}$ sum to $(2^{m+r-1} + 2^j) + (D - 2^j) = D + 2^{m+r-1}$. Hence $L_r = (D + 2^{m+r-1})L_{r-1}$ and $L_0 = 1$. Dividing by $Z = D\prod_{i=0}^{t-2}(D+2^i)$ gives $c_r/(D\prod_{i=0}^{t-r-2}(D+2^i)) = F_{t-r}$. □

Theorem 10. *For all $a, b \geq 0$ with $t = a + b \geq 1$ and all $n \geq 1$,*

$$\lambda_{\max}\big(X^{(a,b)}\big) = F_{t-w}, \qquad w = \min(a,b) + \kappa(|a-b|).$$

Proof. By Lemma 5 it suffices to show $\rho(M) = F_{t-w}$. Levels are dimensions of totally isotropic subspaces, so every level is at most w (Proposition 6); level w is attained by U_N of Lemma 8 for a maximal isotropic N, since $N_L(U_N) = N$. Since $F_{s+1}/F_s = (2^{s-1}+1)/(D + 2^{s-1}) \leq 1$, F_{t-r} is nondecreasing in r, so by Lemmas 7–9 every column sum of M is at most F_{t-w}, and $\rho(M) \leq F_{t-w}$. If $(0,z) \in U'$ then $(0,z) \in U \circ U'$, because $(0,0) \in U$; so $N_R(U \circ U') \supseteq N_R(U')$ and the level never decreases under left multiplication (recall $\dim N_L = \dim N_R$ for every element of Σ, Section 3). Ordering levels increasingly, M is block lower-triangular, and the columns at level w send all of their mass into level w. The diagonal block M_{ww} therefore has every column sum equal to F_{t-w}, so $\mathbf{1}^\mathsf{T} M_{ww} = F_{t-w}\mathbf{1}^\mathsf{T}$ and F_{t-w} is an eigenvalue of M_{ww}, hence of M. Thus $\rho(M) = F_{t-w}$. □

Corollary 11. *For $1 \leq T \leq 3$, $k \geq 1$ and $n \geq 1$,*

$$P_{\mathrm{stab}}(n; T, k) = \binom{D+T-1}{T} F_{T+k-\kappa(k)}.$$

Proof. Lemma 1 and Theorem 10 with $(a,b) = (T+k, T)$, $t = 2T+k$ and $w = T + \kappa(k)$. □

6 Consequences

When the universal cloner is optimal. Let $m = T + k$. Since $\kappa(k) \geq 0$ and F_s is decreasing in s, Corollary 11 gives $P_{\mathrm{stab}}/D_T = F_{m-\kappa(k)} \geq F_m$. The frame-potential inequality gives $F_m \geq 1/D_m$,

with equality exactly when the ensemble is a projective m-design; stabilizer states are 3-designs but not 4-designs [5, Cor. 1], so $F_m > 1/D_m$ for every $m \geq 4$. Conversely, when $m \leq 3$ we have $\kappa(k) = 0$ and $F_m = 1/D_m$. Therefore the universal cloner is optimal exactly when $T + k \leq 3$, in agreement with the design criterion of [7]; the first strict gap appears at $T + k = 4$ (e.g. $1/8$ against $4/35$ at $n = 2$, $1 \to 4$).

Plateaus. The optimum is unchanged from k to $k + 1$ whenever $\kappa(k + 1) = \kappa(k) + 1$, i.e. for $k + 1 \equiv 4, 6, 7, 0 \pmod 8$: for instance $1 \to 4$ and $1 \to 5$ have the same value, as do $1 \to 6$ through $1 \to 9$. Producing one more copy is free exactly when a longer doubly-even code becomes available.

Many copies. For $s \geq n + 1$, cancelling powers of two gives the exact identity

$$F_s = \frac{1}{N_n} \prod_{i=s-1-n}^{s-2} \left(1 + 2^{-i}\right),$$

so $F_s \to 1/N_n$ and $P_{\mathrm{stab}}(n; T, k) \to D_T/N_n$ as $k \to \infty$. For $T \leq 3$ the T-copy states form a tight frame for Sym^T (3-design property), so D_T/N_n is the optimal probability of identifying the state from T copies, attained by the pretty-good measurement (any POVM $\{E_S\}$ succeeds with probability $\frac{1}{N_n} \sum_S \langle S^{\otimes T} | E_S | S^{\otimes T} \rangle \leq \frac{1}{N_n} \operatorname{tr} \sum_S E_S = D_T/N_n$). Cloning many copies thus becomes identification.

Scope. The results concern average global fidelity; by Lemma 1 they are also the worst-case optima over stabilizer states. Theorem 10 holds for every signature, but the reduction to cloning uses $T \leq 3$; for $T \geq 4$ the Clifford group is not a unitary 4-design [6] and the optimal dual need not be scalar.

7 Discussion

The exact optimum interpolates between Werner's universal value (few copies) and optimal identification (many copies), with the transition controlled by doubly-even codes. Natural next questions are the cloning value for $T \geq 4$, an explicit optimal channel, other figures of merit such as the trace distance used in [2], and qudit stabilizer states, where the relevant forms and codes change. Related results concern exact stabilizer identification [9], efficient $O(n)$-copy learning [12] and sample-optimal bounds [13], a sharp copy-rate threshold for fixed-target forgery of quadratic-stabilizer public keys over odd-prime fields [14], and the structure of the Clifford commutant [6, 8, 15, 16].

Verification. The results are proved analytically and do not depend on computer-assisted certification; the following checks are supplementary. Exact symbolic certificates (the spectrum of the orbit quotient of M, valid for every n) were computed for the 15 signatures (a, b), $a \geq b$, with $3 \leq t \leq 7$ other than the trivial $(3, 3)$, and reproduced with the same code on two further machines. Lemmas 7 and 8 were verified exactly over all 9,845,550 relations for each of the five signatures with $t = 8$. A separate implementation written by a second system (its own enumeration of Σ, composition and matrix M) confirmed Lemmas 7 and 8 and $\rho(M) = F_{t-w}$ for the eight signatures with $a \geq b$ and $t \leq 5$ at $n = 1, 2, 3$, and an exhaustive search confirmed Proposition 6 for all 54 signatures with $t \leq 9$. The cloning values agree with direct numerical diagonalization of the stabilizer Gram kernels for $n \leq 3$, exact Gram-matrix certificates at $n = 3$, and Lanczos computations at $n = 4$.

Use of AI tools

This work used several AI systems; the author directed it and takes full responsibility for its content. **Grok Bot** (SpaceXAI) proposed the closed form, proved Theorem 10 with Lemmas 7–9 and the upper bound of Proposition 6, verified the reduction of Lemma 1, and produced the exact certificates and the $t = 8$ check. **Claude** (Anthropic; Claude Opus 5.5) derived the reduction of Lemma 1, wrote the proof of Lemma 3, audited every proof, wrote the separate implementation and the independent numerical checks, reproduced the certificates, and drafted the manuscript. **ChatGPT** (OpenAI) refereed the proofs, finding a false operator identity and an overstated domain in earlier versions, and supplied the short proof of the spectral inclusion and the attainment argument in Lemma 5, the code constructions in Proposition 6, and the choice of w in Lemma 8. **Perplexity** and **Consensus** carried out the prior-art searches.

Data and code availability

Supporting code, computational certificates, logs and reproduction instructions are archived with this preprint's Zenodo record (file `paper3_code.zip`) and in the folder `paper3/` of `https://github.com/Weldog/sidon-phase-cloning` (release tag `paper3-v1`).

Competing interests

The author declares no competing interests. This work received no external funding.

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